

Chapter 4: Fractions and percentages

12 lessons · one lesson for each exercise · each lesson is one hour

NSW syllabus outcome

Stage 4 Number and algebra: MA4-FRC-C-01 represents and operates with fractions, decimals and percentages to solve problems.

CHAPTER

4

4.01 Fractions

4.02 Ordering fractions

4.03 Adding and subtracting fractions

4.04 Adding and subtracting mixed numerals

4.05 Fraction of a quantity

4.06 Multiplying fractions

4.07 Dividing fractions

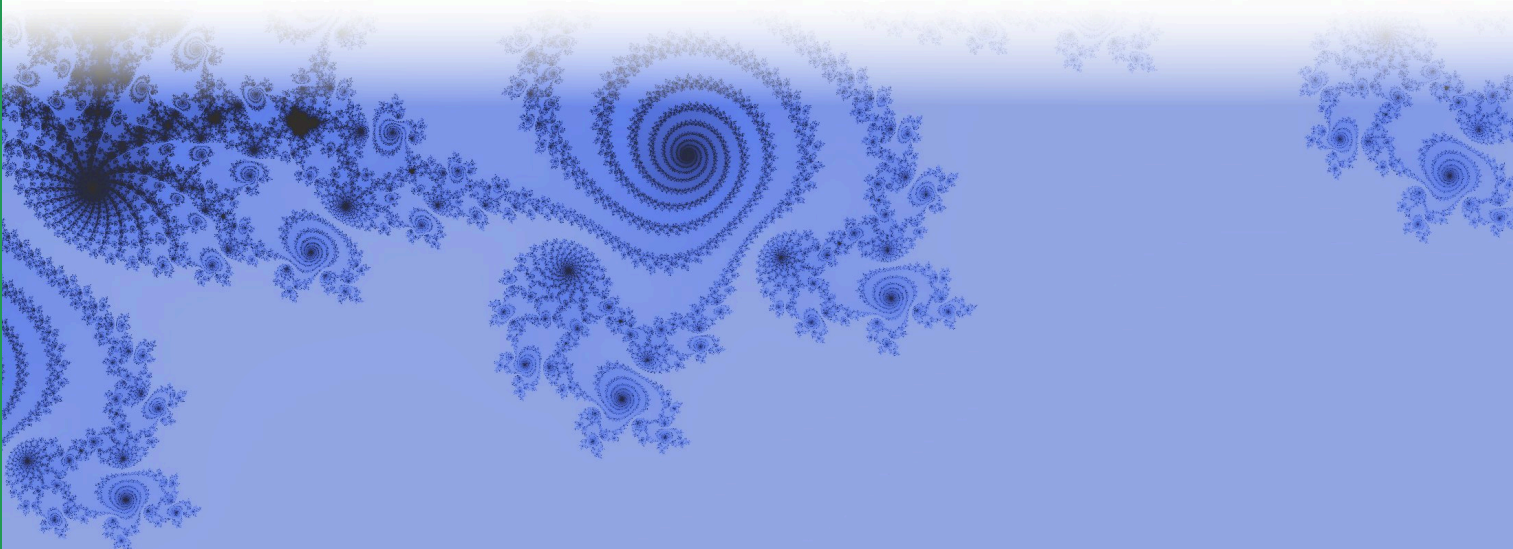
4.08 Fraction problems

4.09 Fractions on a calculator

4.10 Percentages, fractions and decimals

4.11 Percentage of a quantity

4.12 Expressing as a fraction or percentage



How to use this booklet

1 Each lesson is **one hour**. Start with the **learning intentions** and **success criteria**, write the **key terms** and use the **notes** box.

WE DO

Grey boxes. Your teacher works each example (parts a, b, c) on the board. **Copy every step** into the space.

2 Every lesson has three levels: **Easy**, then **Medium**, then **Challenging**. Work down the page: the questions get harder as you go.

YOU DO

Blue boxes. Now it is your turn. Work on your own. Quick drill answers go on the line; show working in the boxes.

DRILL

Skill drill pages. After the Easy and the Medium examples there is a full page of short practice questions. Work down the page, write answers on the lines, then mark it and record your score and time.

3 Finished early? Try the **extension**. At the end, **show off your skill** on the tear-off exit ticket and hand it in.

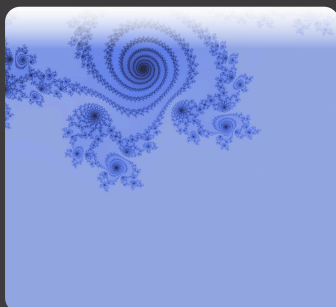
4 Missed a lesson? Each lesson ends with a **summary** with a worked example for every level. Start there.

Contents and progress tracker

Lesson	Topic	Page	Date	Done
4.01	Fractions	3		☐
4.02	Ordering fractions	7		☐
4.03	Adding and subtracting fractions	11		☐
4.04	Adding and subtracting mixed numerals	15		☐
4.05	Fraction of a quantity	19		☐
4.06	Multiplying fractions	23		☐
4.07	Dividing fractions	27		☐
4.08	Fraction problems	31		☐
4.09	Fractions on a calculator	37		☐
4.10	Percentages, fractions and decimals	41		☐
4.11	Percentage of a quantity	45		☐
4.12	Expressing as a fraction or percentage	49		☐

In this chapter you will

- write equivalent fractions and simplify
- order fractions
- add, subtract, multiply and divide fractions and mixed numerals
- find a fraction or percentage of a quantity
- solve fraction problems
- convert between fractions, decimals and percentages



About the artwork

The pictures in this booklet are parts of the **Mandelbrot set**, drawn by a computer that repeats one simple rule, $z \rightarrow z^2 + c$, thousands of times for every point. Every booklet shows a different part of the set: this one is a triple spiral on the edge of the top bulb. Maths can make beautiful things.

Acknowledgement of Country

We acknowledge the Bundjalung people, the Traditional Custodians of the land on which Kingscliff High School stands, and pay our respects to Elders past and present.

LESSON
4.01

Fractions

Date _____

Learning intentions

We are learning to:

- convert between mixed numerals and improper fractions
- write equivalent fractions and simplify fractions

Success criteria

I can:

- ☐ change a mixed numeral to an improper fraction and back
- ☐ multiply top and bottom by the same number to find an equivalent fraction
- ☐ divide top and bottom by the HCF to simplify

Key terms Write the definition as your teacher explains it.

Improper fraction

Mixed numeral

Equivalent

Notes

Level 1 · Getting started

Easy · 32 questions

WE DO With your teacher: copy what they write on the board.

Example 1 Convert between mixed numerals and improper fractions.

a Write $1\frac{3}{4}$ as an improper fraction.

b Write $\frac{7}{4}$ as a mixed numeral.

Your teacher works this out. Copy every step.

Your teacher works this out. Copy every step.

YOU DO On your own: **Easy** practise the skill. Work down the page, one round at a time. Write each answer on its line. **32 QUESTIONS**

REMEMBER Mixed → improper: whole × bottom + top.

ROUND 1 Questions 1–16: simplify each fraction.

1 $\frac{5}{10}$

2 $\frac{66}{72}$

3 $\frac{2}{6}$

4 $\frac{3}{9}$

5 $\frac{6}{12}$

6 $\frac{3}{6}$

7 $\frac{4}{48}$

8 $\frac{12}{40}$

9 $\frac{10}{16}$

10 $\frac{6}{18}$

11 $\frac{55}{60}$

12 $\frac{54}{60}$

13 $\frac{6}{9}$

14 $\frac{15}{25}$

15 $\frac{4}{40}$

16 $\frac{10}{12}$

YOU DO

On your own: **Easy** keep going: Round 2.

16 QUESTIONS MORE

ROUND 2

Questions 17–32: change mixed numerals to improper fractions, and improper fractions to mixed numerals.

17 $5\frac{2}{3}$

18 $2\frac{1}{10}$

19 $\frac{7}{2}$

20 $6\frac{1}{6}$

21 $6\frac{2}{5}$

22 $\frac{8}{3}$

23 $5\frac{7}{10}$

24 $\frac{53}{12}$

25 $4\frac{1}{4}$

26 $1\frac{2}{3}$

27 $1\frac{9}{10}$

28 $\frac{19}{3}$

29 $3\frac{3}{5}$

30 $\frac{17}{5}$

31 $\frac{11}{2}$

32 $\frac{3}{2}$



Level 2 · Building up

Medium · 24 questions

WE DO

With your teacher: copy what they write on the board.

Example 2

Complete each pair of equivalent fractions.

a Complete: $\frac{1}{3} = \frac{?}{15}$

b Complete: $\frac{3}{4} = \frac{12}{?}$

Your teacher works this out. Copy every step.

Your teacher works this out. Copy every step.

YOU DO

On your own: **Medium** practise the skill. Work down the page, one round at a time. Write each answer on its line.

24 QUESTIONS

REMEMBER Improper → mixed: divide top by bottom.

ROUND 1

Questions 1–16: find the missing number in each pair of equivalent fractions.

1 $\frac{5}{8} = \frac{30}{?}$

2 $\frac{3}{4} = \frac{21}{?}$

3 $\frac{2}{5} = \frac{8}{?}$

4 $\frac{1}{6} = \frac{?}{54}$

5 $\frac{1}{2} = \frac{?}{8}$

6 $\frac{6}{10} = \frac{24}{?}$

7 $\frac{4}{10} = \frac{?}{30}$

8 $\frac{3}{4} = \frac{15}{?}$

9 $\frac{1}{2} = \frac{?}{4}$

10 $\frac{3}{4} = \frac{6}{?}$

11 $\frac{3}{6} = \frac{?}{48}$

12 $\frac{8}{12} = \frac{40}{?}$

13 $\frac{1}{2} = \frac{?}{10}$

14 $\frac{1}{3} = \frac{?}{9}$

15 $\frac{1}{3} = \frac{8}{?}$

16 $\frac{2}{3} = \frac{?}{9}$

YOU DO On your own: **Medium** keep going: Round 2.

8 QUESTIONS MORE

ROUND 2 Questions 17–24: simplify fully.

17 $\frac{45}{90}$

18 $\frac{120}{180}$

19 $\frac{120}{135}$

20 $\frac{18}{27}$

21 $\frac{18}{60}$

22 $\frac{12}{180}$

23 $\frac{48}{80}$

24 $\frac{16}{72}$

Level 3 · Challenge

Challenging · 3 questions · then try the extension

WE DO With your teacher: copy what they write on the board.

Example 3 Simplify each fraction.

a Simplify $\frac{25}{30}$.

Your teacher works this out. Copy every step.

b Simplify $\frac{36}{24}$ and write as a mixed numeral.

Your teacher works this out. Copy every step.

YOU DO On your own: **Challenging** Simplify each fraction.

3 QUESTIONS

REMEMBER Equivalent: $\times$ or $\div$ top and bottom by the same number.

1 $\frac{25}{30}$

2 $\frac{30}{40}$

3 $\frac{15}{20}$

✂ Show off your skill: tear along this line and hand it to your teacher ✂

YOU DO **Show off your skill**

Choose **ONE** question. Show your working.

Name _____ Class _____

Level 1

Write $3\frac{2}{5}$ as an improper fraction.

Level 2

Complete: $\frac{2}{7} = \frac{?}{35}$

Level 3

Simplify $\frac{45}{60}$.

Extension · Stretch yourself**Worked example**

Find three fractions equivalent to $\frac{2}{3}$ with denominators between 10 and 20.

1. Multiply top and bottom by the same number
2. $\times 4$, $\times 5$, $\times 6$

Answer: $\frac{8}{12}$, $\frac{10}{15}$, $\frac{12}{18}$

YOU DO

On your own: read the worked example, then try these.

- E1 Which fraction is not equivalent to $\frac{3}{5}$: $\frac{9}{15}$, $\frac{12}{20}$, $\frac{15}{30}$, $\frac{30}{50}$?

- E2 A fraction simplifies to $\frac{2}{7}$. Its numerator and denominator add to 45. Find the fraction.

Lesson summary

Missed the lesson? Start here.

Remember

1. Mixed $\rightarrow$ improper: whole $\times$ bottom + top.
2. Improper $\rightarrow$ mixed: divide top by bottom.
3. Equivalent: $\times$ or $\div$ top and bottom by the same number.
4. Simplify: $\div$ top and bottom by the HCF.

Level 1

$$1\frac{3}{4}$$

1. $1 \times 4 + 3 = 7$

Answer: $\frac{7}{4}$

Level 2

$$\frac{26}{3}$$

1. $26 \div 3 = 8 \text{ r } 2$

Answer: $8\frac{2}{3}$

Level 3

$$\text{Simplify } \frac{24}{42}$$

1. HCF is 6

Answer: $\frac{4}{7}$

Extension

$$\text{Equivalent to } \frac{2}{3}$$

1. $\times 4$, $\times 5$, $\times 6$

Answer: $\frac{8}{12}$, $\frac{10}{15}$, $\frac{12}{18}$

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LESSON
4.02

Ordering fractions

Date _____

Learning intentions

We are learning to:

- compare and order fractions

Success criteria

I can:

- ☐ compare fractions with the same denominator
- ☐ change fractions to a common denominator
- ☐ order fractions from smallest to largest

Key terms Write the definition as your teacher explains it.

Common denominator

Ascending order

Notes

Level 1 · Getting started

Easy · 28 questions

WE DO With your teacher: copy what they write on the board.

Example 1 Which fraction is larger?

a Which is larger: $\frac{3}{8}$ or $\frac{5}{8}$?

b Which is larger: $\frac{2}{3}$ or $\frac{3}{5}$?

Your teacher works this out. Copy every step.

Your teacher works this out. Copy every step.

YOU DO On your own: **Easy** practise the skill. Work down the page, one round at a time. Write each answer on its line. **28 QUESTIONS**

REMEMBER Same denominator: compare the numerators.

ROUND 1 Questions 1–16: which fraction is larger? The denominators (or numerators) are the same.

1 $\frac{2}{4}$ or $\frac{3}{4}$

2 $\frac{7}{10}$ or $\frac{5}{10}$

3 $\frac{1}{5}$ or $\frac{1}{8}$

4 $\frac{6}{10}$ or $\frac{5}{10}$

5 $\frac{5}{6}$ or $\frac{5}{12}$

6 $\frac{6}{10}$ or $\frac{9}{10}$

7 $\frac{11}{12}$ or $\frac{2}{12}$

8 $\frac{4}{11}$ or $\frac{4}{9}$

9 $\frac{4}{12}$ or $\frac{4}{10}$

10 $\frac{1}{10}$ or $\frac{1}{2}$

11 $\frac{2}{5}$ or $\frac{4}{5}$

12 $\frac{3}{11}$ or $\frac{3}{7}$

13 $\frac{9}{10}$ or $\frac{8}{10}$

14 $\frac{2}{4}$ or $\frac{2}{11}$

15 $\frac{1}{3}$ or $\frac{1}{6}$

16 $\frac{2}{7}$ or $\frac{1}{7}$

YOU DO

On your own: **Easy** keep going: Round 2.

12 QUESTIONS MORE

ROUND 2

Questions 17–28: write both fractions with the lowest common denominator.

17 $\frac{1}{2}$ and $\frac{7}{8}$

18 $\frac{2}{3}$ and $\frac{6}{12}$

19 $\frac{7}{10}$ and $\frac{3}{6}$

20 $\frac{3}{5}$ and $\frac{1}{3}$

21 $\frac{1}{6}$ and $\frac{2}{5}$

22 $\frac{3}{4}$ and $\frac{1}{12}$

23 $\frac{3}{6}$ and $\frac{2}{3}$

24 $\frac{6}{12}$ and $\frac{9}{10}$

25 $\frac{3}{8}$ and $\frac{9}{10}$

26 $\frac{11}{12}$ and $\frac{4}{5}$

27 $\frac{6}{10}$ and $\frac{1}{2}$

28 $\frac{1}{3}$ and $\frac{1}{6}$



Level 2 · Building up

Medium · 20 questions

WE DO

With your teacher: copy what they write on the board.

Example 2

Order from smallest to largest.

a Order $\frac{4}{5}$, $\frac{3}{10}$, $\frac{2}{5}$, $\frac{7}{10}$ from smallest to largest.

Your teacher works this out. Copy every step.

b Order $\frac{1}{2}$, $\frac{1}{3}$, $\frac{7}{12}$, $\frac{11}{12}$.

Your teacher works this out. Copy every step.

YOU DO

On your own: **Medium** practise the skill. Work down the page, one round at a time. Write each answer on its line.

20 QUESTIONS

REMEMBER Different denominators: find the LCM of the denominators.

ROUND 1

Questions 1–16: which is larger? Use a common denominator.

1 $\frac{9}{12}$ or $\frac{1}{2}$

2 $\frac{1}{2}$ or $\frac{7}{8}$

3 $\frac{3}{4}$ or $\frac{1}{6}$

4 $\frac{1}{10}$ or $\frac{2}{3}$

5 $\frac{1}{4}$ or $\frac{1}{2}$

6 $\frac{8}{10}$ or $\frac{1}{12}$

7 $\frac{3}{4}$ or $\frac{1}{3}$

8 $\frac{4}{5}$ or $\frac{1}{2}$

9 $\frac{7}{10}$ or $\frac{1}{4}$

10 $\frac{9}{10}$ or $\frac{2}{3}$

11 $\frac{4}{6}$ or $\frac{10}{12}$

12 $\frac{1}{3}$ or $\frac{1}{2}$

13 $\frac{5}{12}$ or $\frac{4}{10}$

14 $\frac{4}{5}$ or $\frac{2}{3}$

15 $\frac{3}{10}$ or $\frac{2}{5}$

16 $\frac{5}{6}$ or $\frac{2}{12}$

YOU DO On your own: **Medium** keep going: Round 2.

4 QUESTIONS MORE

ROUND 2 Questions 17–20: write in ascending order (smallest first).

17 $\frac{2}{4}, \frac{8}{12}, \frac{3}{4}$

18 $\frac{10}{12}, \frac{1}{4}, \frac{4}{12}$

19 $\frac{3}{6}, \frac{2}{5}, \frac{1}{8}$

20 $\frac{1}{3}, \frac{1}{2}, \frac{4}{10}$

Level 3 · Challenge

Challenging · 3 questions · then try the extension

WE DO With your teacher: copy what they write on the board.

Example 3 Order from smallest to largest.

a Order $\frac{3}{4}, \frac{2}{3}, \frac{5}{6}, \frac{7}{12}$.

Your teacher works this out. Copy every step.

b Order $1\frac{1}{3}, \frac{5}{4}, \frac{11}{8}$.

Your teacher works this out. Copy every step.

YOU DO On your own: **Challenging** Order from smallest to largest.

3 QUESTIONS

REMEMBER Rewrite each fraction with that denominator.

1 $\frac{3}{4}, \frac{2}{3}, \frac{5}{6}, \frac{7}{12}$

2 $\frac{3}{5}, \frac{1}{2}, \frac{7}{10}, \frac{4}{15}$

3 $1\frac{1}{3}, \frac{5}{4}, \frac{11}{8}, 1\frac{1}{6}$

✕ Show off your skill: tear along this line and hand it to your teacher ✕

YOU DO **Show off your skill**

Choose **ONE** question. Show your working.

Name _____ Class _____

Level 1

Which is larger: $\frac{4}{9}$ or $\frac{7}{9}$?

Level 2

Which is larger: $\frac{3}{4}$ or $\frac{4}{5}$?

Level 3

Order $\frac{5}{6}, \frac{2}{3}, \frac{3}{4}$.

Extension · Stretch yourself**Worked example**

Order $\frac{5}{7}$, $\frac{2}{3}$, $\frac{7}{10}$ from smallest to largest.

1. Common denominator 210

2. $\frac{150}{210}$, $\frac{140}{210}$, $\frac{147}{210}$

Answer: $\frac{2}{3}$, $\frac{7}{10}$, $\frac{5}{7}$

YOU DO

On your own: read the worked example, then try these.

E1 Order $\frac{4}{9}$, $\frac{3}{7}$, $\frac{1}{2}$.

E2 Find a fraction exactly halfway between $\frac{1}{4}$ and $\frac{1}{3}$.

Lesson summary

Missed the lesson? Start here.

- Remember**
1. Same denominator: compare the numerators.
 2. Different denominators: find the LCM of the denominators.
 3. Rewrite each fraction with that denominator.
 4. Order the numerators.

Level 1

$\frac{3}{8}$ or $\frac{5}{8}$

1. Same denominator

Answer: $\frac{5}{8}$

Level 2

$\frac{2}{3}$ or $\frac{3}{5}$

1. $\frac{10}{15}$ vs $\frac{9}{15}$

Answer: $\frac{2}{3}$

Level 3

$\frac{1}{2}$, $\frac{1}{3}$, $\frac{7}{12}$, $\frac{11}{12}$

1. Twelfths: 6, 4, 7, 11

Answer: $\frac{1}{3}$, $\frac{1}{2}$, $\frac{7}{12}$, $\frac{11}{12}$

Extension

$\frac{5}{7}$, $\frac{2}{3}$, $\frac{7}{10}$

1. Denominator 210

Answer: $\frac{2}{3}$, $\frac{7}{10}$, $\frac{5}{7}$

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LESSON
4.03

Adding and subtracting fractions

Date _____

Learning intentions

We are learning to:

- add and subtract fractions

Success criteria

I can:

- ☐ add and subtract fractions with the same denominator
- ☐ use a common denominator when the denominators are different
- ☐ simplify the answer

Key terms Write the definition as your teacher explains it.

Numerator

Denominator

Notes

Level 1 · Getting started

Easy · 32 questions

WE DO

With your teacher: copy what they write on the board.

Example 1

Evaluate. The denominators are the same.

a $\frac{3}{4} + \frac{1}{4}$

b $\frac{5}{7} - \frac{2}{7}$

Your teacher works this out. Copy every step.

Your teacher works this out. Copy every step.

YOU DO

On your own: **Easy** practise the skill. Work down the page, one round at a time. Write each answer on its line.

32 QUESTIONS

REMEMBER Same denominator: add or subtract the numerators.

ROUND 1

Questions 1–16: evaluate. The denominators are the same. Simplify your answer.

1 $\frac{2}{3} + \frac{1}{3}$

2 $\frac{5}{6} + \frac{3}{6}$

3 $\frac{3}{4} + \frac{1}{4}$

4 $\frac{4}{6} + \frac{2}{6}$

5 $\frac{3}{4} - \frac{1}{4}$

6 $\frac{3}{5} + \frac{4}{5}$

7 $\frac{10}{11} - \frac{2}{11}$

8 $\frac{4}{5} - \frac{2}{5}$

9 $\frac{6}{7} + \frac{5}{7}$

10 $\frac{9}{11} - \frac{7}{11}$

11 $\frac{11}{12} - \frac{10}{12}$

12 $\frac{6}{8} - \frac{4}{8}$

13 $\frac{1}{12} + \frac{4}{12}$

14 $\frac{9}{10} + \frac{8}{10}$

15 $\frac{5}{7} + \frac{6}{7}$

16 $\frac{2}{4} + \frac{1}{4}$

YOU DO

On your own: **Easy** keep going: Round 2.

16 QUESTIONS MORE

ROUND 2

Questions 17–32: evaluate. Change one fraction so the denominators match.

17 $\frac{2}{3} + \frac{6}{9}$

18 $\frac{1}{3} + \frac{2}{6}$

19 $\frac{1}{2} + \frac{2}{4}$

20 $\frac{1}{2} - \frac{1}{4}$

21 $\frac{2}{3} + \frac{1}{6}$

22 $\frac{4}{5} - \frac{1}{10}$

23 $\frac{2}{5} + \frac{10}{15}$

24 $\frac{4}{5} - \frac{4}{15}$

25 $\frac{3}{4} - \frac{2}{12}$

26 $\frac{1}{2} + \frac{3}{4}$

27 $\frac{1}{5} + \frac{9}{15}$

28 $\frac{1}{3} + \frac{4}{6}$

29 $\frac{2}{4} - \frac{3}{8}$

30 $\frac{1}{4} + \frac{6}{8}$

31 $\frac{3}{5} + \frac{6}{15}$

32 $\frac{2}{5} + \frac{5}{10}$



Level 2 · Building up

Medium · 22 questions

WE DO

With your teacher: copy what they write on the board.

Example 2

Evaluate. Use a common denominator.

a $\frac{2}{5} + \frac{3}{10}$

b $\frac{3}{4} - \frac{1}{5}$

Your teacher works this out. Copy every step.

Your teacher works this out. Copy every step.

YOU DO

On your own: **Medium** practise the skill. Work down the page, one round at a time. Write each answer on its line.

22 QUESTIONS

REMEMBER Different: find the lowest common denominator.

ROUND 1

Questions 1–16: evaluate. Use the lowest common denominator.

1 $\frac{1}{3} + \frac{7}{10}$

2 $\frac{3}{6} + \frac{1}{3}$

3 $\frac{2}{6} + \frac{1}{2}$

4 $\frac{1}{2} - \frac{1}{8}$

5 $\frac{7}{12} + \frac{2}{4}$

6 $\frac{6}{8} - \frac{1}{2}$

7 $\frac{9}{10} + \frac{3}{6}$

8 $\frac{8}{10} + \frac{2}{3}$

9 $\frac{2}{4} + \frac{2}{3}$

10 $\frac{6}{10} + \frac{5}{6}$

11 $\frac{4}{5} - \frac{1}{2}$

12 $\frac{4}{8} + \frac{1}{12}$

13 $\frac{1}{3} + \frac{2}{8}$

14 $\frac{1}{4} + \frac{3}{8}$

15 $\frac{4}{10} + \frac{1}{8}$

16 $\frac{1}{2} - \frac{2}{5}$

YOU DO On your own: **Medium** keep going: Round 2.

6 QUESTIONS MORE

ROUND 2 Questions 17–22: evaluate.

17 $\frac{1}{2} + \frac{1}{3} - \frac{1}{8}$

18 $\frac{1}{3} + \frac{1}{6} - \frac{1}{6}$

19 $\frac{1}{4} + \frac{1}{3} - \frac{1}{12}$

20 $\frac{1}{2} + \frac{1}{3} - \frac{1}{12}$

21 $\frac{1}{3} + \frac{1}{4} - \frac{1}{12}$

22 $\frac{1}{2} + \frac{1}{4} - \frac{1}{12}$

Level 3 · Challenge

Challenging · 3 questions · then try the extension

WE DO With your teacher: copy what they write on the board.

Example 3 Evaluate. Use the lowest common denominator.

a $\frac{1}{2} + \frac{1}{3}$

b $\frac{5}{6} + \frac{3}{4}$

Your teacher works this out. Copy every step.

Your teacher works this out. Copy every step.

YOU DO On your own: **Challenging** Evaluate.

3 QUESTIONS

REMEMBER Rewrite, then add or subtract.

1 $\frac{5}{6} + \frac{4}{5}$

2 $\frac{1}{2} + \frac{1}{3} + \frac{1}{4}$

3 $\frac{7}{8} - \frac{1}{3} - \frac{1}{6}$

✕ Show off your skill: tear along this line and hand it to your teacher ✕

YOU DO **Show off your skill** Choose **ONE** question. Show your working.

Name _____ Class _____

Level 1

$\frac{2}{7} + \frac{3}{7}$

Level 2

$\frac{1}{2} - \frac{1}{6}$

Level 3

$\frac{2}{3} + \frac{3}{5}$

Extension · Stretch yourself**Worked example**Find the missing fraction: $\frac{3}{8} + ? = \frac{5}{6}$

1. $?\ = \frac{5}{6} - \frac{3}{8}$

2. $\ = \frac{20}{24} - \frac{9}{24}$

Answer: $\frac{11}{24}$ **YOU DO**

On your own: read the worked example, then try these.

E1 Find the missing fraction: $? - \frac{2}{5} = \frac{1}{3}$

E2 Write $\frac{7}{12}$ as the sum of two different unit fractions (numerator 1).

Lesson summary

Missed the lesson? Start here.

Remember 1. Same denominator: add or subtract the numerators.
3. Rewrite, then add or subtract.

2. Different: find the lowest common denominator.
4. Simplify the answer.

Level 1

$\frac{3}{4} + \frac{1}{4}$

1. $\frac{4}{4}$

Answer: 1

Level 2

$\frac{2}{5} + \frac{3}{10}$

1. $\frac{4}{10} + \frac{3}{10}$

Answer: $\frac{7}{10}$ **Level 3**

$\frac{3}{4} - \frac{1}{5}$

1. $\frac{15}{20} - \frac{4}{20}$

Answer: $\frac{11}{20}$ **Extension**

$\frac{3}{8} + ? = \frac{5}{6}$

1. $\frac{20}{24} - \frac{9}{24}$

Answer: $\frac{11}{24}$

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LESSON
4.04

Adding and subtracting mixed numerals

Date _____

Learning intentions

We are learning to:

- add and subtract mixed numerals

Success criteria

I can:

- ☐ add the whole numbers, then the fractions
- ☐ change mixed numerals to improper fractions when subtracting needs it
- ☐ write the answer as a simplified mixed numeral

Key terms Write the definition as your teacher explains it.

Mixed numeral

Whole number

Notes

Level 1 · Getting started

Easy · 32 questions

WE DO With your teacher: copy what they write on the board.

Example 1 Evaluate. Add the whole numbers, then the fractions.

a $2\frac{1}{5} + 6\frac{1}{2}$

b $3\frac{3}{4} + 1\frac{1}{2}$

Your teacher works this out. Copy every step.

Your teacher works this out. Copy every step.

YOU DO On your own: **Easy** practise the skill. Work down the page, one round at a time. Write each answer on its line. **32 QUESTIONS**

REMEMBER Adding: add wholes, then add fractions.

ROUND 1 Questions 1–16: evaluate. Add (or subtract) the whole numbers, then the fractions.

1 $8\frac{3}{4} - 6\frac{1}{4}$

2 $8\frac{2}{3} + 4\frac{1}{3}$

3 $5\frac{1}{4} + 3\frac{3}{4}$

4 $3\frac{2}{4} + 2\frac{1}{4}$

5 $7\frac{8}{10} - 2\frac{6}{10}$

6 $5\frac{1}{4} - 3\frac{1}{4}$

7 $3\frac{3}{5} + 2\frac{3}{5}$

8 $3\frac{7}{9} + 2\frac{5}{9}$

9 $7\frac{1}{3} + 5\frac{1}{3}$

10 $3\frac{3}{5} + 1\frac{2}{5}$

11 $6\frac{6}{10} + 4\frac{5}{10}$

12 $8\frac{6}{8} - 2\frac{2}{8}$

13 $7\frac{1}{7} - 2\frac{1}{7}$

14 $8\frac{5}{6} - 6\frac{3}{6}$

15 $7\frac{2}{3} + 6\frac{1}{3}$

16 $7\frac{1}{4} - 4\frac{1}{4}$

YOU DO

On your own: **Easy** keep going: Round 2.

16 QUESTIONS MORE

ROUND 2

Questions 17–32: evaluate. Take the fraction from one whole.

17 $8 - \frac{1}{9}$

18 $5 - \frac{5}{6}$

19 $4 - \frac{3}{6}$

20 $9 - \frac{1}{4}$

21 $5 - \frac{4}{5}$

22 $8 - \frac{5}{8}$

23 $2 - \frac{9}{10}$

24 $5 - \frac{2}{3}$

25 $7 - \frac{6}{7}$

26 $7 - \frac{3}{5}$

27 $7 - \frac{2}{3}$

28 $7 - \frac{1}{4}$

29 $3 - \frac{2}{3}$

30 $7 - \frac{5}{9}$

31 $2 - \frac{1}{2}$

32 $3 - \frac{1}{5}$



Level 2 · Building up

Medium · 18 questions

WE DO

With your teacher: copy what they write on the board.

Example 2

Evaluate.

a $10 - \frac{1}{3}$

Your teacher works this out. Copy every step.

b $3\frac{2}{3} - 2\frac{1}{6}$

Your teacher works this out. Copy every step.

YOU DO

On your own: **Medium** practise the skill. Work down the page, one round at a time. Write each answer on its line.

18 QUESTIONS

REMEMBER Carry if the fraction is more than 1.

ROUND 1

Questions 1–12: evaluate. Use a common denominator.

1 $4\frac{1}{2} + 1\frac{2}{5}$

2 $5\frac{1}{2} + 5\frac{3}{4}$

3 $3\frac{1}{5} + 5\frac{1}{2}$

4 $3\frac{1}{2} + 1\frac{1}{8}$

5 $4\frac{1}{3} + 3\frac{5}{12}$

6 $5\frac{9}{10} + 1\frac{2}{3}$

7 $6\frac{1}{8} + 1\frac{1}{3}$

8 $3\frac{3}{5} + 3\frac{1}{3}$

9 $4\frac{5}{12} + 4\frac{1}{2}$

10 $2\frac{5}{8} + 5\frac{1}{3}$

11 $2\frac{3}{4} + 4\frac{7}{8}$

12 $3\frac{7}{12} + 4\frac{2}{3}$

YOU DO On your own: **Medium** keep going: Round 2.

6 QUESTIONS MORE

ROUND 2 Questions 13–18: evaluate. You may need to change to improper fractions.

13 $3\frac{3}{8} - 1\frac{7}{8}$

14 $8\frac{1}{5} - 1\frac{2}{5}$

15 $5\frac{1}{4} - 2\frac{3}{4}$

16 $7\frac{1}{5} - 2\frac{4}{5}$

17 $8\frac{1}{3} - 3\frac{2}{3}$

18 $6\frac{1}{3} - 2\frac{2}{3}$

Level 3 · Challenge

Challenging · 3 questions · then try the extension

WE DO With your teacher: copy what they write on the board.

Example 3 Evaluate. Change to improper fractions.

a $5\frac{1}{4} - 3\frac{2}{3}$

b $2\frac{3}{4} - 1\frac{3}{5} + 1\frac{1}{2}$

Your teacher works this out. Copy every step.

Your teacher works this out. Copy every step.

YOU DO On your own: **Challenging** Evaluate.

3 QUESTIONS

REMEMBER Subtracting: change to improper fractions if needed.

1 $5\frac{1}{4} - 3\frac{2}{3}$

2 $3\frac{1}{2} - 1\frac{4}{5}$

3 $2\frac{1}{2} + 1\frac{1}{3} + 2\frac{3}{4}$

✂ Show off your skill: tear along this line and hand it to your teacher ✂

YOU DO **Show off your skill** Choose **ONE** question. Show your working.

Name _____ Class _____

Level 1

$2\frac{1}{3} + 4\frac{1}{3}$

Level 2

$6 - \frac{2}{5}$

Level 3

$4\frac{1}{3} - 2\frac{3}{4}$

Extension · Stretch yourself**Worked example**Evaluate $2\frac{3}{4} - 1\frac{3}{5} + 1\frac{1}{2}$.

1. Improper: $\frac{11}{4} - \frac{8}{5} + \frac{3}{2}$
2. Denominator 20: $\frac{55}{20} - \frac{32}{20} + \frac{30}{20}$
3. $\frac{53}{20}$

Answer: $2\frac{13}{20}$ **YOU DO**

On your own: read the worked example, then try these.

E1 Evaluate $5\frac{1}{3} - 2\frac{1}{2} - 1\frac{3}{4}$.

E2 Two mixed numerals add to $7\frac{1}{6}$. One is $3\frac{2}{3}$. Find the other.

Lesson summary

Missed the lesson? Start here.

- Remember**
1. Adding: add wholes, then add fractions.
 2. Carry if the fraction is more than 1.
 3. Subtracting: change to improper fractions if needed.
 4. Write the answer as a mixed numeral.

Level 1

$2\frac{1}{5} + 6\frac{1}{2}$

1. $8 + \frac{7}{10}$

Answer: $8\frac{7}{10}$ **Level 2**

$10 - \frac{1}{3}$

1. $9\frac{3}{3} - \frac{1}{3}$

Answer: $9\frac{2}{3}$ **Level 3**

$5\frac{1}{4} - 3\frac{2}{3}$

1. $\frac{63}{12} - \frac{44}{12}$

Answer: $1\frac{7}{12}$ **Extension**

$2\frac{3}{4} - 1\frac{3}{5} + 1\frac{1}{2}$

1. $\frac{53}{20}$

Answer: $2\frac{13}{20}$

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LESSON
4.05

Fraction of a quantity

Date _____

Learning intentions

We are learning to:

- find a fraction of a quantity, including with units

Success criteria

I can:

- ☐ divide by the denominator, then multiply by the numerator
- ☐ convert units before finding the fraction
- ☐ give the answer in the unit asked for

Key terms Write the definition as your teacher explains it.

Quantity

Unit

Notes

Level 1 · Getting started

Easy · 32 questions

WE DO With your teacher: copy what they write on the board.

Example 1 Evaluate.

a $\frac{1}{4}$ of 28

b $\frac{3}{7}$ of 21

Your teacher works this out. Copy every step.

Your teacher works this out. Copy every step.

YOU DO On your own: **Easy** practise the skill. Work down the page, one round at a time. Write each answer on its line. **32 QUESTIONS**

REMEMBER “of” means multiply.

ROUND 1 Questions 1–16: evaluate. Divide by the denominator.

1 $\frac{1}{10}$ of 60

2 $\frac{1}{6}$ of 18

3 $\frac{1}{2}$ of 12

4 $\frac{1}{10}$ of 120

5 $\frac{1}{6}$ of 12

6 $\frac{1}{5}$ of 15

7 $\frac{1}{7}$ of 21

8 $\frac{1}{2}$ of 16

9 $\frac{1}{12}$ of 36

10 $\frac{1}{2}$ of 14

11 $\frac{1}{12}$ of 96

12 $\frac{1}{7}$ of 77

13 $\frac{1}{6}$ of 54

14 $\frac{1}{4}$ of 36

15 $\frac{1}{3}$ of 6

16 $\frac{1}{5}$ of 20

YOU DO

On your own: **Easy** keep going: Round 2.

16 QUESTIONS MORE

ROUND 2

Questions 17–32: evaluate. Divide by the denominator, then multiply by the numerator.

17 $\frac{5}{7}$ of 70

18 $\frac{7}{10}$ of 20

19 $\frac{9}{10}$ of 80

20 $\frac{5}{6}$ of 18

21 $\frac{3}{10}$ of 70

22 $\frac{4}{5}$ of 20

23 $\frac{4}{5}$ of 45

24 $\frac{4}{7}$ of 56

25 $\frac{3}{5}$ of 50

26 $\frac{7}{9}$ of 90

27 $\frac{3}{7}$ of 77

28 $\frac{2}{3}$ of 27

29 $\frac{2}{5}$ of 40

30 $\frac{4}{9}$ of 99

31 $\frac{4}{5}$ of 50

32 $\frac{7}{10}$ of 30



Level 2 · Building up

Medium · 20 questions

WE DO

With your teacher: copy what they write on the board.

Example 2

Evaluate in the unit given.

a $\frac{2}{3}$ of 1 day (in hours)

Your teacher works this out. Copy every step.

b $\frac{4}{5}$ of 2 km (in metres)

Your teacher works this out. Copy every step.

YOU DO

On your own: **Medium** practise the skill. Work down the page, one round at a time. Write each answer on its line.

20 QUESTIONS

REMEMBER Divide by the denominator.

ROUND 1

Questions 1–12: evaluate in the unit given.

1 $\frac{19}{25}$ of 1 kg (g)

2 $\frac{1}{2}$ of 1 kg (g)

3 $\frac{3}{4}$ of a day (h)

4 $\frac{1}{20}$ of 1 kg (g)

5 $\frac{3}{5}$ of an hour (min)

6 $\frac{1}{5}$ of 1 m (cm)

7 $\frac{3}{8}$ of 1 km (m)

8 $\frac{1}{2}$ of an hour (min)

9 $\frac{2}{3}$ of a day (h)

10 $\frac{1}{2}$ of a day (h)

11 $\frac{1}{4}$ of 1 km (m)

12 $\frac{1}{4}$ of 1 kg (g)

YOU DO On your own: **Medium** keep going: Round 2.

8 QUESTIONS MORE

ROUND 2 Questions 13–20: what fraction is the first amount of the second? Simplify.

13 24 of 30

14 4 of 20

15 10 of 24

16 10 of 15

17 2 of 12

18 12 of 60

19 8 of 15

20 13 of 24

Level 3 · Challenge

Challenging · 3 questions · then try the extension

WE DO With your teacher: copy what they write on the board.

Example 3 Evaluate in the unit given.

a $\frac{3}{4}$ of 3 hours (in minutes)

Your teacher works this out. Copy every step.

b $\frac{1}{6}$ of 5 minutes (in seconds)

Your teacher works this out. Copy every step.

YOU DO On your own: **Challenging** Word problems.

3 QUESTIONS

REMEMBER Multiply by the numerator.

1 $\frac{3}{5}$ of a class of 30 are girls. How many boys are there?

2 A phone costs \$840. Mia has saved $\frac{5}{7}$ of the price. How much more does she need?

3 $\frac{2}{3}$ of a number is 24. Find the number.

✂ Show off your skill: tear along this line and hand it to your teacher ✂

YOU DO **Show off your skill** Choose **ONE** question. Show your working.

Name _____ Class _____

Level 1

$\frac{1}{5}$ of 45

Level 2

$\frac{5}{6}$ of 72

Level 3

$\frac{3}{4}$ of 2 hours (in minutes)

Extension · Stretch yourself**Worked example**

After spending $\frac{1}{4}$ of his money and then $\frac{1}{3}$ of what was left, Ali has \$30. How much did he start with?

1. After $\frac{1}{4}$: $\frac{3}{4}$ left
2. Then $\frac{2}{3}$ of $\frac{3}{4} = \frac{1}{2}$ of the money is left
3. $\frac{1}{2} = \$30$

Answer: \$60**YOU DO**

On your own: read the worked example, then try these.

- E1** $\frac{2}{5}$ of a number is 18. What is $\frac{3}{4}$ of the number?

- E2** A tank is $\frac{3}{8}$ full. After adding 60 L it is $\frac{3}{4}$ full. How much does it hold?

Lesson summary

Missed the lesson? Start here.

Remember 1. “of” means multiply.
3. Multiply by the numerator.

2. Divide by the denominator.
4. Change units first if needed.

Level 1 $\frac{3}{7}$ of 21

1. $21 \div 7 = 3$
2. 3×3

Answer: 9**Level 2** $\frac{4}{5}$ of 2 km in m

1. 2 km = 2000 m
2. $2000 \div 5 \times 4$

Answer: 1600 m**Level 3** $\frac{3}{4}$ of 3 h in min

1. $180 \div 4 \times 3$

Answer: 135 min**Extension**

Ali's money

1. $\frac{1}{2}$ left is \$30

Answer: \$60

This space is left blank on purpose. It is the back of your exit ticket.

LESSON
4.06

Multiplying fractions

Date _____

Learning intentions

We are learning to:

- multiply fractions and mixed numerals

Success criteria

I can:

- ☐ multiply numerators and multiply denominators
- ☐ cancel common factors first
- ☐ change mixed numerals to improper fractions first

Key terms Write the definition as your teacher explains it.

Product

Cancel

Notes

Level 1 · Getting started

Easy · 32 questions

WE DO With your teacher: copy what they write on the board.

Example 1 Evaluate.

a $\frac{2}{3} \times \frac{1}{5}$

b $10 \times \frac{3}{5}$

Your teacher works this out. Copy every step.

Your teacher works this out. Copy every step.

YOU DO On your own: **Easy** practise the skill. Work down the page, one round at a time. Write each answer on its line. **32 QUESTIONS**

REMEMBER Change mixed numerals to improper fractions.

ROUND 1 Questions 1–16: evaluate. Multiply the numerators and the denominators. Simplify.

1 $\frac{5}{9} \times \frac{2}{8}$

2 $\frac{1}{2} \times \frac{2}{9}$

3 $\frac{1}{5} \times \frac{6}{7}$

4 $\frac{7}{8} \times \frac{4}{8}$

5 $\frac{1}{5} \times \frac{4}{5}$

6 $\frac{1}{4} \times \frac{2}{9}$

7 $\frac{1}{3} \times \frac{3}{4}$

8 $\frac{5}{7} \times \frac{1}{8}$

9 $\frac{2}{7} \times \frac{7}{8}$

10 $\frac{1}{5} \times \frac{1}{2}$

11 $\frac{4}{5} \times \frac{3}{6}$

12 $\frac{3}{5} \times \frac{1}{2}$

13 $\frac{3}{5} \times \frac{2}{3}$

14 $\frac{2}{8} \times \frac{5}{9}$

15 $\frac{8}{9} \times \frac{1}{4}$

16 $\frac{1}{3} \times \frac{1}{8}$

YOU DO

On your own: **Easy** keep going: Round 2.

16 QUESTIONS MORE

ROUND 2

Questions 17–32: evaluate.

17 $4 \times \frac{1}{4}$

18 $\frac{1}{3} \times 6$

19 $\frac{6}{10} \times 4$

20 $\frac{1}{5} \times 8$

21 $\frac{6}{10} \times 12$

22 $\frac{5}{7} \times 11$

23 $\frac{2}{3} \times 9$

24 $\frac{4}{6} \times 5$

25 $2 \times \frac{1}{2}$

26 $11 \times \frac{2}{4}$

27 $11 \times \frac{1}{4}$

28 $\frac{2}{10} \times 6$

29 $5 \times \frac{2}{4}$

30 $10 \times \frac{1}{7}$

31 $5 \times \frac{4}{5}$

32 $\frac{4}{9} \times 7$



Level 2 · Building up

Medium · 22 questions

WE DO

With your teacher: copy what they write on the board.

Example 2

Evaluate. Cancel first.

a $\frac{4}{7} \times \frac{5}{8}$

b $\frac{9}{16} \times \frac{2}{45}$

Your teacher works this out. Copy every step.

Your teacher works this out. Copy every step.

YOU DO

On your own: **Medium** practise the skill. Work down the page, one round at a time. Write each answer on its line.

22 QUESTIONS

REMEMBER Cancel common factors (top with bottom).

ROUND 1

Questions 1–16: evaluate. Cancel first.

1 $\frac{6}{12} \times \frac{3}{10}$

2 $\frac{20}{35} \times \frac{5}{16}$

3 $\frac{9}{28} \times \frac{8}{12}$

4 $\frac{20}{30} \times \frac{24}{35}$

5 $\frac{12}{30} \times \frac{5}{21}$

6 $\frac{12}{18} \times \frac{18}{20}$

7 $\frac{4}{20} \times \frac{4}{12}$

8 $\frac{5}{20} \times \frac{10}{15}$

9 $\frac{15}{35} \times \frac{5}{12}$

10 $\frac{12}{16} \times \frac{4}{42}$

11 $\frac{4}{42} \times \frac{12}{14}$

12 $\frac{12}{16} \times \frac{4}{18}$

13 $\frac{6}{12} \times \frac{8}{12}$

14 $\frac{6}{18} \times \frac{9}{15}$

15 $\frac{16}{42} \times \frac{6}{20}$

16 $\frac{6}{10} \times \frac{8}{36}$

YOU DO On your own: **Medium** keep going: Round 2.

6 QUESTIONS MORE

ROUND 2 Questions 17–22: evaluate. Change to improper fractions first.

17 $3\frac{1}{2} \times 3\frac{2}{3}$

18 $1\frac{1}{3} \times 3\frac{4}{5}$

19 $2\frac{1}{3} \times 1\frac{2}{3}$

20 $1\frac{1}{2} \times 2\frac{3}{5}$

21 $1\frac{1}{2} \times 2\frac{2}{3}$

22 $1\frac{2}{3} \times 1\frac{2}{3}$

Level 3 · Challenge

Challenging · 3 questions · then try the extension

WE DO With your teacher: copy what they write on the board.

Example 3 Evaluate. Change to improper fractions first.

a $1\frac{1}{2} \times \frac{1}{2}$

Your teacher works this out. Copy every step.

b $2\frac{1}{3} \times 4\frac{2}{3}$

Your teacher works this out. Copy every step.

YOU DO On your own: **Challenging** Evaluate.

3 QUESTIONS

REMEMBER Multiply tops, multiply bottoms.

1 $2\frac{1}{3} \times 4\frac{2}{3}$

2 $3\frac{2}{4} \times \frac{2}{3}$

3 $2\frac{1}{2} \times 1\frac{2}{5} \times \frac{1}{3}$

✂ Show off your skill: tear along this line and hand it to your teacher ✂

YOU DO **Show off your skill** Choose **ONE** question. Show your working.

Name _____ Class _____

Level 1

$\frac{3}{5} \times \frac{1}{4}$

Level 2

$\frac{5}{9} \times \frac{3}{10}$

Level 3

$1\frac{1}{4} \times 2\frac{2}{3}$

Extension · Stretch yourself**Worked example**Evaluate $\frac{2}{3} \times \frac{3}{4} \times \frac{4}{5} \times \frac{5}{6}$.

1. Cancel: 3, 4 and 5 cancel
2. Leaves $\frac{2}{6}$

Answer: $\frac{1}{3}$ **YOU DO**

On your own: read the worked example, then try these.

E1 Evaluate $\frac{1}{2} \times \frac{2}{3} \times \frac{3}{4} \times \dots \times \frac{9}{10}$.**E2** A rectangle is $3\frac{1}{2}$ m by $2\frac{2}{3}$ m. Find its area.**Lesson summary**

Missed the lesson? Start here.

- Remember**
1. Change mixed numerals to improper fractions.
 2. Cancel common factors (top with bottom).
 3. Multiply tops, multiply bottoms.
 4. Write as a simplified mixed numeral.

Level 1

$$\frac{2}{3} \times \frac{1}{5}$$

1. 2×1 and 3×5

Answer: $\frac{2}{15}$ **Level 2**

$$\frac{4}{7} \times \frac{5}{8}$$

1. Cancel 4 and 8

Answer: $\frac{5}{14}$ **Level 3**

$$2\frac{1}{3} \times 4\frac{2}{3}$$

$$1. \frac{7}{3} \times \frac{14}{3}$$

Answer: $10\frac{8}{9}$ **Extension**

Telescoping product

1. Cancel

Answer: $\frac{1}{3}$

This space is left blank on purpose. It is the back of your exit ticket.

LESSON
4.07

Dividing fractions

Date _____

Learning intentions

We are learning to:

- divide fractions and mixed numerals

Success criteria

I can:

- ☐ find the **reciprocal** of a fraction
- ☐ divide by multiplying by the reciprocal
- ☐ change mixed numerals to improper fractions first

Key terms Write the definition as your teacher explains it.

Reciprocal

Quotient

Notes

Level 1 · Getting started

Easy · 32 questions

WE DO With your teacher: copy what they write on the board.

Example 1 Write the reciprocal, then divide.

a Write the reciprocal of $\frac{3}{4}$, 5 and $2\frac{1}{2}$.

b $10 \div \frac{1}{5}$

Your teacher works this out. Copy every step.

Your teacher works this out. Copy every step.

YOU DO On your own: **Easy** practise the skill. Work down the page, one round at a time. Write each answer on its line.

32 QUESTIONS

REMEMBER Change mixed numerals to improper fractions.

ROUND 1 Questions 1–16: write the reciprocal.

1 $\frac{10}{2}$

2 5

3 $\frac{4}{3}$

4 $\frac{5}{10}$

5 $\frac{4}{7}$

6 4

7 8

8 $\frac{10}{4}$

9 $\frac{1}{6}$

10 $\frac{7}{8}$

11 3

12 6

13 $\frac{5}{7}$

14 $\frac{8}{2}$

15 $\frac{4}{6}$

16 $\frac{3}{12}$

YOU DO

On your own: **Easy** keep going: Round 2.

16 QUESTIONS MORE

ROUND 2

Questions 17–32: evaluate. How many parts fit into the whole number?

17 $6 \div \frac{1}{2}$

18 $2 \div \frac{1}{7}$

19 $2 \div \frac{1}{6}$

20 $6 \div \frac{1}{6}$

21 $8 \div \frac{1}{5}$

22 $5 \div \frac{1}{3}$

23 $6 \div \frac{1}{7}$

24 $7 \div \frac{1}{7}$

25 $4 \div \frac{1}{5}$

26 $4 \div \frac{1}{3}$

27 $5 \div \frac{1}{5}$

28 $9 \div \frac{1}{2}$

29 $3 \div \frac{1}{8}$

30 $2 \div \frac{1}{8}$

31 $7 \div \frac{1}{3}$

32 $8 \div \frac{1}{6}$



Level 2 · Building up

Medium · 22 questions

WE DO

With your teacher: copy what they write on the board.

Example 2

Evaluate. Multiply by the reciprocal.

a $\frac{1}{2} \div 3$

b $\frac{3}{4} \div \frac{3}{8}$

Your teacher works this out. Copy every step.

Your teacher works this out. Copy every step.

YOU DO

On your own: **Medium** practise the skill. Work down the page, one round at a time. Write each answer on its line.

22 QUESTIONS

REMEMBER Keep the first fraction.

ROUND 1

Questions 1–16: evaluate. Multiply by the reciprocal.

1 $\frac{4}{6} \div \frac{1}{3}$

2 $\frac{1}{3} \div \frac{1}{2}$

3 $\frac{1}{3} \div \frac{6}{7}$

4 $\frac{2}{5} \div \frac{4}{5}$

5 $\frac{2}{4} \div \frac{1}{5}$

6 $\frac{5}{6} \div \frac{1}{4}$

7 $\frac{1}{3} \div \frac{2}{3}$

8 $\frac{1}{2} \div \frac{3}{6}$

9 $\frac{1}{2} \div \frac{1}{5}$

10 $\frac{8}{9} \div \frac{2}{6}$

11 $\frac{3}{8} \div \frac{3}{9}$

12 $\frac{1}{2} \div \frac{2}{9}$

13 $\frac{2}{6} \div \frac{2}{3}$

14 $\frac{1}{7} \div \frac{3}{5}$

15 $\frac{5}{7} \div \frac{6}{8}$

16 $\frac{3}{7} \div \frac{2}{3}$

YOU DO

On your own: **Medium** keep going: Round 2.

6 QUESTIONS MORE

ROUND 2

Questions 17–22: evaluate. Change to improper fractions first.

17 $3\frac{1}{2} \div 2\frac{1}{2}$

18 $2\frac{1}{2} \div 1\frac{2}{5}$

19 $1\frac{2}{4} \div 1\frac{2}{4}$

20 $2\frac{1}{2} \div 1\frac{2}{3}$

21 $2\frac{3}{5} \div 2\frac{1}{2}$

22 $3\frac{1}{5} \div 1\frac{1}{2}$



Level 3 · Challenge

Challenging · 3 questions · then try the extension

WE DO

With your teacher: copy what they write on the board.

Example 3

Evaluate. Change to improper fractions first.

a $2\frac{1}{4} \div 1\frac{1}{8}$

b $1\frac{7}{8} \div 1\frac{1}{4}$

Your teacher works this out. Copy every step.

Your teacher works this out. Copy every step.

YOU DO

On your own: **Challenging** Evaluate.

3 QUESTIONS

REMEMBER

Change $\div$ to $\times$ and flip the second fraction.

① $1\frac{1}{2} \div \frac{1}{2}$

② $2\frac{1}{4} \div 1\frac{1}{8}$

③ $1\frac{7}{8} \div 1\frac{1}{4}$

✂ Show off your skill: tear along this line and hand it to your teacher ✂

YOU DO

Show off your skill

Choose **ONE** question. Show your working.

Name	Class
------	-------



Write the reciprocal of $\frac{5}{8}$.



$$\frac{2}{3} \div \frac{1}{6}$$



$$3\frac{1}{2} \div 1\frac{3}{4}$$

Extension · Stretch yourself

Worked example

A jug holds $2\frac{1}{4}$ L. How many $\frac{3}{10}$ L glasses can be filled?

$$1. \frac{9}{4} \div \frac{3}{10} = \frac{9}{4} \times \frac{10}{3}$$

$$2. = \frac{90}{12} = 7\frac{1}{2}$$

Answer: 7 full glasses

YOU DO

On your own: read the worked example, then try these.

E1 $\frac{2}{3}$ of a number is $\frac{4}{9}$. Find the number.

E2 Evaluate $(\frac{1}{2} + \frac{1}{3}) \div (\frac{1}{2} - \frac{1}{3})$.

Lesson summary

Missed the lesson? Start here.

Remember 1. Change mixed numerals to improper fractions. 2. Keep the first fraction.
3. Change $\div$ to $\times$ and flip the second fraction. 4. Multiply and simplify.

Level 1

$$10 \div \frac{1}{5}$$

$$1. 10 \times 5$$

Answer: 50

Level 2

$$\frac{3}{4} \div \frac{3}{8}$$

$$1. \frac{3}{4} \times \frac{8}{3}$$

Answer: 2

Level 3

$$2\frac{1}{4} \div 1\frac{1}{8}$$

$$1. \frac{9}{4} \times \frac{8}{9}$$

Answer: 2

Extension

Glasses from a jug

$$1. 7\frac{1}{2}$$

Answer: 7 glasses

This space is left blank on purpose. It is the back of your exit ticket.

LESSON
4.08

Fraction problems

Date _____

Learning intentions

We are learning to:

- solve problems involving fractions

Success criteria

I can:

- ☐ decide which operation the problem needs
- ☐ find the “rest” by subtracting fractions from 1
- ☐ check the answer makes sense

Key terms Write the definition as your teacher explains it.

Remainder

Total

Notes

Level 1 · Getting started

Easy · 21 questions

WE DO With your teacher: copy what they write on the board.

Example 1 Read the problem, then answer.

- a** A punch has 2 L ginger ale, $1\frac{1}{2}$ L orange juice, 2 $\frac{3}{4}$ L lemonade and $\frac{1}{3}$ L raspberry juice. How many litres of punch?

Your teacher works this out. Copy every step.

- b** How much more lemonade than orange juice is in the punch?

Your teacher works this out. Copy every step.

YOU DO On your own: **Easy** practise the skill. Work down the page, one round at a time. Write each answer on its line. **21 QUESTIONS**

REMEMBER Read the problem twice. Underline the numbers.

ROUND 1 Questions 1–9: what fraction is left?

- 1** $\frac{3}{5}$ of a garden is planted

- 2** $\frac{6}{8}$ of a pizza is eaten

- 3** $\frac{8}{12}$ of a pizza is eaten

- 4** $\frac{5}{6}$ of a class is away

- 5** $\frac{2}{10}$ of a class is away

- 6** $\frac{2}{6}$ of a class is away

- 7** $\frac{2}{3}$ of a pizza is eaten

- 8** $\frac{6}{10}$ of a pizza is eaten

- 9** $\frac{2}{5}$ of a book is read

YOU DO

On your own: **Easy** keep going: Round 2.

12 QUESTIONS MORE

ROUND 2

Questions 10–21: find the amount.

10 $\frac{1}{2}$ of 12 kilograms

11 $\frac{1}{5}$ of \$15

12 $\frac{1}{5}$ of \$20

13 $\frac{2}{5}$ of 45 kilograms

14 $\frac{1}{10}$ of 100 minutes

15 $\frac{1}{2}$ of \$20

16 $\frac{1}{2}$ of 16 minutes

17 $\frac{1}{2}$ of 18 kilograms

18 $\frac{3}{4}$ of 28 minutes

19 $\frac{7}{10}$ of \$20

20 $\frac{2}{3}$ of \$27

21 $\frac{5}{6}$ of \$60

Level 2 · Building up

Medium · 14 questions

WE DO With your teacher: copy what they write on the board.**Example 2** Read the problem, then answer.

- a** A farmer has 24 goats. $\frac{1}{4}$ are black and $\frac{2}{3}$ are white. What fraction are neither?

Your teacher works this out. Copy every step.

- b** The rest of the goats are split equally between brown and spotted. How many are spotted?

Your teacher works this out. Copy every step.

YOU DO On your own: **Medium** practise the skill. Work down the page, one round at a time. Write each answer on its line (show working in the boxes).

14 QUESTIONS

REMEMBER “Of” → multiply. “Altogether” → add.**ROUND 1** Questions 1–10: how much is left?

- 1** Sam had \$24 and spent $\frac{3}{4}$ of it. How much is left?

- 2** Sam had \$45 and spent $\frac{2}{3}$ of it. How much is left?

- 3** Sam had \$40 and spent $\frac{3}{5}$ of it. How much is left?

- 4** Sam had \$95 and spent $\frac{2}{5}$ of it. How much is left?

- 5** Sam had \$88 and spent $\frac{5}{8}$ of it. How much is left?

- 6** Sam had \$20 and spent $\frac{4}{5}$ of it. How much is left?

- 7** Sam had \$72 and spent $\frac{5}{8}$ of it. How much is left?

- 8** Sam had \$18 and spent $\frac{1}{3}$ of it. How much is left?

- 9** Sam had \$50 and spent $\frac{1}{5}$ of it. How much is left?

- 10** Sam had \$42 and spent $\frac{1}{3}$ of it. How much is left?

YOU DO On your own: **Medium** keep going: Round 2.

4 QUESTIONS MORE

ROUND 2 Questions 11–14: two steps: find each part, then answer.

- 11 60 people: $\frac{1}{2}$ walk, $\frac{1}{5}$ ride. The rest take the bus.
How many take the bus?

- 12 120 people: $\frac{1}{4}$ walk, $\frac{1}{6}$ ride. The rest take the bus.
How many take the bus?

- 13 20 people: $\frac{1}{2}$ walk, $\frac{1}{5}$ ride. The rest take the bus.
How many take the bus?

- 14 24 people: $\frac{1}{3}$ walk, $\frac{1}{4}$ ride. The rest take the bus.
How many take the bus?

Level 3 · Challenge

Challenging · 3 questions · then try the extension

WE DO With your teacher: copy what they write on the board.

Example 3 Read the problem, then answer.

- a Jo reads $\frac{2}{5}$ of a book on Monday and $\frac{1}{3}$ on Tuesday. What fraction is left?

Your teacher works this out. Copy every step.

- b The book has 240 pages. How many pages are left?

Your teacher works this out. Copy every step.

YOU DO On your own: **Challenging** Harder problems.

3 QUESTIONS

REMEMBER “Left” or “the rest” → subtract from 1 (the whole).

- 1 $\frac{3}{8}$ of a school's 480 students catch a bus and $\frac{1}{3}$ of the rest walk. How many walk?

- 2 A rope is cut into pieces of $\frac{2}{5}$ m. There are 12 pieces and $\frac{1}{10}$ m left over. How long was the rope?

- 3 After giving away $\frac{2}{7}$ of her cards, Zoe has 45 left. How many did she start with?



Extension · Stretch yourself

Finished the challenge? Stretch yourself here.



Worked example

A cyclist rides $\frac{1}{3}$ of a trip on day 1 and $\frac{1}{4}$ of the trip on day 2. She has 50 km left. How long is the trip?

1. $\frac{1}{3} + \frac{1}{4} = \frac{7}{12}$
2. $\frac{5}{12}$ of the trip is 50 km
3. $\frac{1}{12}$ is 10 km

Answer: 120 km

YOU DO

On your own: read the worked example, then try these.

- E1 $\frac{1}{4}$ of a class plays soccer, $\frac{1}{3}$ plays netball and the other 10 play neither. How many in the class?

- E2 A bottle is $\frac{2}{3}$ full. After pouring out 150 mL it is $\frac{1}{6}$ full. How much does it hold?

✂ Show off your skill: tear along this line and hand it to your teacher ✂

YOU DO

Show off your skill

Choose ONE question. Show your working.

Name _____ Class _____

Level 1

What is $\frac{2}{5} + \frac{1}{3}$ of a whole?

Level 2

 $\frac{3}{4}$ of 36 students passed. How many failed?

Level 3

After spending $\frac{3}{5}$ of his money, Kai has \$28. How much did he start with?

Lesson summary

Missed the lesson? Start here.

- Remember**
1. Read the problem twice. Underline the numbers.
 2. “Of” → multiply. “Altogether” → add.
 3. “Left” or “the rest” → subtract from 1 (the whole).
 4. Answer the question with units.

Level 1

Punch total

$$1. 2 + 1\frac{1}{2} + 2\frac{3}{4} + \frac{1}{3}$$

Answer: $6\frac{7}{12}$ L**Level 2**

Goats: neither black nor white

$$1. 1 - \frac{1}{4} - \frac{2}{3}$$

Answer: $\frac{1}{12}$ **Level 3**

Spotted goats

1. $\frac{1}{12}$ of 24 = 2
2. $2 \div 2$

Answer: 1 goat

Extension

Cyclist

$$1. \frac{5}{12} = 50 \text{ km}$$

Answer: 120 km

My reflection: what did I learn today? Which question was hardest, and why?

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LESSON
4.09

Fractions on a calculator

Date _____

Learning intentions

We are learning to:

- use a calculator to work with fractions

Success criteria

I can:

- ☐ enter fractions and mixed numerals with the fraction key
- ☐ convert between improper fractions, mixed numerals and decimals
- ☐ check calculator answers with an estimate

Key terms Write the definition as your teacher explains it.

Fraction key

S ⇌ D key

Notes

Level 1 · Getting started

Easy · 32 questions

WE DO With your teacher: copy what they write on the board.

Example 1 Use the fraction key on your calculator.

a Enter $\frac{3}{4} + \frac{1}{6}$.

b Enter $3\frac{1}{2} \times \frac{2}{5}$.

Your teacher works this out. Copy every step.

Your teacher works this out. Copy every step.

YOU DO On your own: **Easy** practise the skill. Work down the page, one round at a time. Write each answer on its line. **32 QUESTIONS**

REMEMBER Use the fraction key to enter a fraction.

ROUND 1 Questions 1–16: use the fraction key on your calculator.

1 $\frac{9}{18} - \frac{4}{11}$

2 $\frac{5}{6} - \frac{1}{8}$

3 $\frac{3}{8} + \frac{3}{12}$

4 $\frac{11}{18} + \frac{9}{13}$

5 $\frac{10}{19} - \frac{5}{13}$

6 $\frac{3}{8} + \frac{1}{5}$

7 $\frac{6}{9} + \frac{1}{5}$

8 $\frac{5}{7} - \frac{1}{12}$

9 $\frac{1}{15} + \frac{11}{16}$

10 $\frac{6}{20} + \frac{2}{19}$

11 $\frac{4}{6} + \frac{6}{7}$

12 $\frac{2}{19} + \frac{1}{9}$

13 $\frac{8}{13} - \frac{5}{20}$

14 $\frac{10}{12} + \frac{3}{18}$

15 $\frac{3}{6} + \frac{8}{14}$

16 $\frac{11}{15} - \frac{4}{9}$

YOU DO On your own: **Easy** keep going: Round 2.

16 QUESTIONS MORE

ROUND 2 Questions 17–32: use your calculator.

17 $\frac{2}{7} \times \frac{2}{5}$

18 $\frac{11}{14} \div \frac{9}{11}$

19 $\frac{2}{5} \times \frac{8}{14}$

20 $\frac{18}{19} \times \frac{15}{17}$

21 $\frac{9}{11} \div \frac{10}{15}$

22 $\frac{3}{14} \times \frac{3}{8}$

23 $\frac{5}{10} \div \frac{4}{9}$

24 $\frac{2}{11} \div \frac{3}{11}$

25 $\frac{19}{20} \div \frac{14}{19}$

26 $\frac{6}{17} \times \frac{9}{11}$

27 $\frac{2}{6} \times \frac{7}{13}$

28 $\frac{1}{5} \times \frac{2}{15}$

29 $\frac{3}{20} \div \frac{14}{16}$

30 $\frac{6}{13} \times \frac{2}{7}$

31 $\frac{3}{17} \div \frac{16}{17}$

32 $\frac{8}{10} \times \frac{8}{18}$

Level 2 · Building up

Medium · 20 questions

WE DO With your teacher: copy what they write on the board.

Example 2 Use your calculator to convert and evaluate.

a Convert $\frac{17}{5}$ to a mixed numeral and to a decimal.

Your teacher works this out. Copy every step.

b Evaluate $\frac{13}{18} - \frac{3}{8}$.

Your teacher works this out. Copy every step.

YOU DO On your own: **Medium** practise the skill. Work down the page, one round at a time. Write each answer on its line.

20 QUESTIONS

REMEMBER Use SHIFT for a mixed numeral.

ROUND 1 Questions 1–12: use your calculator. Use brackets.

1 $(\frac{3}{4} + \frac{3}{4}) \times \frac{4}{9}$

2 $(\frac{6}{8} + \frac{1}{2}) \times \frac{3}{9}$

3 $(\frac{2}{3} + \frac{6}{8}) \times \frac{5}{6}$

4 $(\frac{1}{2} + \frac{2}{7}) \times \frac{5}{7}$

5 $(\frac{1}{7} + \frac{1}{4}) \times \frac{4}{7}$

6 $(\frac{1}{7} + \frac{2}{5}) \times \frac{3}{8}$

7 $(\frac{5}{7} + \frac{3}{7}) \times \frac{1}{3}$

8 $(\frac{5}{6} + \frac{1}{2}) \times \frac{2}{8}$

9 $(\frac{3}{8} + \frac{4}{8}) \times \frac{5}{7}$

10 $(\frac{1}{4} + \frac{7}{9}) \times \frac{1}{4}$

11 $(\frac{4}{7} + \frac{1}{4}) \times \frac{1}{2}$

12 $(\frac{4}{8} + \frac{2}{3}) \times \frac{7}{9}$

YOU DO On your own: **Medium** keep going: Round 2.

8 QUESTIONS MORE

ROUND 2 Questions 13–20: use your calculator to write each as a mixed numeral.

13 $\frac{96}{11}$

14 $\frac{11}{3}$

15 $\frac{18}{13}$

16 $\frac{79}{12}$

17 $\frac{28}{15}$

18 $\frac{124}{15}$

19 $\frac{41}{9}$

20 $\frac{30}{11}$

Level 3 · Challenge

Challenging · 3 questions · then try the extension

WE DO With your teacher: copy what they write on the board.

Example 3 Use your calculator. Use brackets.

a Evaluate $4\frac{2}{3} \div \frac{1}{3} - 1\frac{3}{4}$.

b Evaluate $2\frac{1}{4}^2$.

Your teacher works this out. Copy every step.

Your teacher works this out. Copy every step.

YOU DO On your own: **Challenging** Use a calculator.

3 QUESTIONS

REMEMBER S $\leftrightarrow$ D switches between fraction and decimal.

1 $4\frac{2}{3} \div \frac{1}{3} - 1\frac{3}{4}$

2 $2\frac{1}{4}^2 + \frac{1}{2}$

3 $(\frac{3}{5} + \frac{1}{4}) \times 2\frac{2}{3}$

✕ Show off your skill: tear along this line and hand it to your teacher ✕

YOU DO **Show off your skill** Choose **ONE** question. Show your working.

Name _____ Class _____

Level 1

$\frac{2}{7} + \frac{3}{5}$

Level 2

$3\frac{4}{5} \times \frac{5}{6}$

Level 3

$2\frac{5}{6} \div \frac{2}{3} - \frac{1}{4}$

Extension · Stretch yourself**Worked example**

Find the fraction halfway between $\frac{5}{9}$ and $\frac{7}{11}$.

1. Add and divide by 2

2. $(\frac{5}{9} + \frac{7}{11}) \div 2$

Answer: $\frac{59}{99}$

YOU DO

On your own: read the worked example, then try these.

E1 Evaluate $\frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \frac{1}{16} + \frac{1}{32}$. What do you notice?

E2 Which is closer to 1: $\frac{11}{13}$ or $\frac{13}{15}$?

Lesson summary

Missed the lesson? Start here.

- Remember**
1. Use the fraction key to enter a fraction.
 2. Use SHIFT for a mixed numeral.
 3. S $\leftrightarrow$ D switches between fraction and decimal.
 4. Estimate first to check the answer.

Level 1

$$\frac{3}{4} + \frac{1}{6}$$

1. Enter with the fraction key

Answer: $\frac{11}{12}$

Level 2

$$\frac{17}{5}$$

1. Convert to mixed

Answer: $3\frac{2}{5} = 3.4$

Level 3

$$4\frac{2}{3} \div \frac{1}{3} - 1\frac{3}{4}$$

1. Use brackets if unsure

Answer: $12\frac{1}{4}$

Extension

Halfway

1. Add and halve

Answer: $\frac{59}{99}$

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LESSON
4.10

Percentages, fractions and decimals

Date _____

Learning intentions

We are learning to:

- convert between percentages, fractions and decimals

Success criteria

I can:

- ☐ write a percentage as a fraction out of 100 and simplify
- ☐ write a percentage as a decimal by dividing by 100
- ☐ write a fraction or decimal as a percentage by multiplying by 100%

Key terms Write the definition as your teacher explains it.

Percentage

Per cent

Notes

Level 1 · Getting started

Easy · 32 questions

WE DO With your teacher: copy what they write on the board.

Example 1 Write as a fraction and a decimal.

a Write 13% as a fraction and a decimal.

Your teacher works this out. Copy every step.

b Write 8% as a fraction and a decimal.

Your teacher works this out. Copy every step.

YOU DO On your own: **Easy** practise the skill. Work down the page, one round at a time. Write each answer on its line. **32 QUESTIONS**

REMEMBER % → fraction: write over 100, simplify.

ROUND 1 Questions 1–16: write each percentage as a fraction in simplest form.

- | | | | |
|---------------|---------------|---------------|---------------|
| 1 25% | 2 75% | 3 35% | 4 8% |
| _____ | _____ | _____ | _____ |
| 5 45% | 6 1% | 7 80% | 8 15% |
| _____ | _____ | _____ | _____ |
| 9 40% | 10 12% | 11 90% | 12 60% |
| _____ | _____ | _____ | _____ |
| 13 20% | 14 10% | 15 4% | 16 70% |
| _____ | _____ | _____ | _____ |

YOU DO On your own: **Medium** keep going: Round 2.

6 QUESTIONS MORE

ROUND 2 Questions 17–22: write each decimal as a percentage and as a fraction.

17 0.3

18 0.4

19 0.2

20 0.12

21 0.5

22 0.75

Level 3 · Challenge

Challenging · 3 questions · then try the extension

WE DO With your teacher: copy what they write on the board.

Example 3 Write as a percentage.

a Write $\frac{23}{25}$ as a percentage.

Your teacher works this out. Copy every step.

b Write 0.045 as a percentage.

Your teacher works this out. Copy every step.

YOU DO On your own: **Challenging** Complete the conversions.

3 QUESTIONS

REMEMBER Fraction or decimal → %: $\times 100\%$.

1 Write $\frac{3}{8}$ as a decimal and a percentage.

2 Write $\frac{1}{3}$ as a percentage.

3 Write $12\frac{1}{2}\%$ as a fraction.

✕ Show off your skill: tear along this line and hand it to your teacher ✕

YOU DO **Show off your skill** Choose **ONE** question. Show your working.

Name _____ Class _____

Level 1

Write 25% as a fraction.

Level 2

Write 0.07 as a percentage.

Level 3

Write $\frac{3}{16}$ as a percentage.

Extension · Stretch yourself**Worked example**

Write $0.\overline{3}$ ($0.333\dots$) as a percentage and as a fraction.

- $0.333\dots \times 100\% = 33.3\dots\%$
- $= 33\frac{1}{3}\%$
- As a fraction: $\frac{1}{3}$

Answer: $33\frac{1}{3}\% = \frac{1}{3}$

YOU DO

On your own: read the worked example, then try these.

- E1 Write $\frac{2}{3}$ as a percentage.

- E2 Write $12\frac{1}{3}\%$ as a fraction in simplest form.

Lesson summary

Missed the lesson? Start here.

- Remember**
- $\% \rightarrow$ fraction: write over 100, simplify.
 - Fraction or decimal $\rightarrow \%$: $\times 100\%$.

- $\% \rightarrow$ decimal: $\div 100$ (move 2 places left).
- Learn: $\frac{1}{2} = 50\%$, $\frac{1}{4} = 25\%$, $\frac{1}{5} = 20\%$, $\frac{1}{10} = 10\%$.

Level 1

13%

1. $\frac{13}{100}$

Answer: $\frac{13}{100} = 0.13$

Level 2

$7\frac{1}{2}\%$

1. $\frac{15}{2} \div 100 = \frac{15}{200}$

Answer: $\frac{3}{40} = 0.075$

Level 3

$\frac{23}{25}$

1. $\frac{92}{100}$

Answer: 92%

Extension

$\frac{1}{3}$

1. $\times 100\% = \frac{100}{3}\%$

Answer: $33\frac{1}{3}\%$

This space is left blank on purpose. It is the back of your exit ticket.

LESSON
4.11

Percentage of a quantity

Date _____

Learning intentions

We are learning to:

- find a percentage of a quantity

Success criteria

I can:

- ☐ change the percentage to a fraction or decimal and multiply
- ☐ use 10% and 1% to find percentages mentally
- ☐ solve problems with percentages

Key terms Write the definition as your teacher explains it.

Discount

Percentage

Notes

Level 1 · Getting started

Easy · 32 questions

WE DO With your teacher: copy what they write on the board.

Example 1 Find mentally.

a Find 10% of 350.

b Find 25% of 84.

Your teacher works this out. Copy every step.

Your teacher works this out. Copy every step.

YOU DO On your own: **Easy** practise the skill. Work down the page, one round at a time. Write each answer on its line. **32 QUESTIONS**

REMEMBER Change % to a fraction or decimal.

ROUND 1 Questions 1–16: find mentally.

1 25% of 112

2 10% of 520

3 50% of 168

4 25% of 68

5 10% of 250

6 25% of 44

7 25% of 116

8 50% of 28

9 1% of 200

10 10% of 360

11 10% of 140

12 1% of 600

13 50% of 52

14 1% of 400

15 50% of 118

16 50% of 92

YOU DO

On your own: **Easy** keep going: Round 2.

16 QUESTIONS MORE

ROUND 2

Questions 17–32: find mentally. 5% is half of 10%. 20% is $2 \times 10\%$.

- | | | | | | | | |
|-------|------------|-------|------------|-------|------------|-------|------------|
| 17 | 20% of 120 | 18 | 20% of 360 | 19 | 30% of 260 | 20 | 20% of 100 |
| <hr/> | | <hr/> | | <hr/> | | <hr/> | |
| 21 | 20% of 560 | 22 | 5% of 280 | 23 | 5% of 580 | 24 | 5% of 520 |
| <hr/> | | <hr/> | | <hr/> | | <hr/> | |
| 25 | 75% of 60 | 26 | 30% of 520 | 27 | 30% of 600 | 28 | 5% of 340 |
| <hr/> | | <hr/> | | <hr/> | | <hr/> | |
| 29 | 20% of 320 | 30 | 20% of 40 | 31 | 75% of 580 | 32 | 5% of 260 |
| <hr/> | | <hr/> | | <hr/> | | <hr/> | |

 Level 2 · Building up

Medium · 22 questions

WE DO

With your teacher: copy what they write on the board.

Example 2

Find the percentage of the quantity.

- a** Find 15% of \$60.

Your teacher works this out. Copy every step.

- b** Find 7% of 1200.

Your teacher works this out. Copy every step.

YOU DO

On your own: **Medium** practise the skill. Work down the page, one round at a time. Write each answer on its line.

22 QUESTIONS

REMEMBER

Multiply by the quantity.

ROUND 1

Questions 1–16: find the percentage of the amount.

- | | | | |
|----------------|-----------------|-----------------|----------------|
| 1 60% of \$120 | 2 10% of \$360 | 3 85% of \$250 | 4 20% of \$330 |
| 5 60% of \$310 | 6 40% of \$220 | 7 95% of \$100 | 8 25% of \$290 |
| 9 15% of \$330 | 10 10% of \$230 | 11 75% of \$110 | 12 70% of \$50 |
| 13 50% of \$30 | 14 70% of \$370 | 15 90% of \$350 | 16 75% of \$70 |

YOU DO

On your own: **Medium** keep going: Round 2.

6 QUESTIONS MORE

ROUND 2

Questions 17–22: find the sale price.

17 \$130, 25% off

18 \$220, 10% off

19 \$260, 10% off

20 \$60, 10% off

21 \$40, 50% off

22 \$160, 30% off



Level 3 · Challenge

Challenging · 3 questions · then try the extension

WE DO

With your teacher: copy what they write on the board.

Example 3

Find, then solve the problem.

a Find 12.5% of 480 kg.

b A \$250 jacket is 30% off. Find the sale price.

Your teacher works this out. Copy every step.

Your teacher works this out. Copy every step.

YOU DO

On your own: **Challenging** Percentage problems.

3 QUESTIONS

REMEMBER 10%: $\div 10$. 1%: $\div 100$. Build others from these.

1 A \$250 jacket is 30% off. Find the sale price.

2 64% of 25 students caught a bus. How many did not?

3 A \$40 meal has a 15% tip added. Find the total.

✂ Show off your skill: tear along this line and hand it to your teacher ✂

YOU DO

Show off your skill

Choose **ONE** question. Show your working.

Name	Class
John	Math
Jane	Math
John	Science
Jane	Science
John	History
Jane	History
John	Art
Jane	Art
John	Music
Jane	Music
John	Physical Education
Jane	Physical Education



Find 10% of 80.



Find 35% of 200.



A \$60 shirt is 15% off. Find the sale price.

Extension · Stretch yourself**Worked example**

A price rises by 10%, then falls by 10%. Is it back to the original price?

1. Try $\$100: + 10\% = \110
2. $- 10\% \text{ of } \$110 = \$110 - \$11 = \99

Answer: No: it is 1% lower

YOU DO

On your own: read the worked example, then try these.

- E1** A \$80 item has 20% off, then another 10% off the new price. Find the final price.

- E2** Is 20% off then 10% off the same as 30% off? Explain.

Lesson summary

Missed the lesson? Start here.

- Remember**
1. Change % to a fraction or decimal.
 2. Multiply by the quantity.
 3. $10\%: \div 10$. $1\%: \div 100$. Build others from these.
 4. Discount: subtract from the price.

Level 1

10% of 350

1. $350 \div 10$

Answer: 35

Level 2

15% of \$60

1. $10\% = \$6$, $5\% = \$3$

Answer: \$9

Level 3

30% off \$250

1. $30\% \text{ of } 250 = 75$
2. $250 - 75$

Answer: \$175

Extension

+10% then -10%

1. $100 \rightarrow 110 \rightarrow 99$

Answer: No, 1% less

This space is left blank on purpose. It is the back of your exit ticket.

LESSON
4.12

Expressing as a fraction or percentage

Date _____

Learning intentions

We are learning to:

- express one quantity as a fraction or percentage of another

Success criteria

I can:

- ☐ write the part over the whole
- ☐ convert both quantities to the same unit first
- ☐ multiply by 100% to get a percentage

Key terms Write the definition as your teacher explains it.

Part

Whole

Notes

Level 1 · Getting started

Easy · 32 questions

WE DO With your teacher: copy what they write on the board.

Example 1 Write as a fraction and a percentage.

- a** Write 51 out of 60 as a fraction and a percentage.

Your teacher works this out. Copy every step.

- b** Write 18 out of 24 as a percentage.

Your teacher works this out. Copy every step.

YOU DO On your own: **Easy** practise the skill. Work down the page, one round at a time. Write each answer on its line. **32 QUESTIONS**

REMEMBER Write **PART** ÷ **WHOLE** as a fraction.

ROUND 1 Questions 1–16: write as a percentage.

- | | | | |
|--------------------------------|---------------------------------|---------------------------------|---------------------------------|
| 1 2 out of 20
_____ | 2 5 out of 10
_____ | 3 3 out of 4
_____ | 4 2 out of 4
_____ |
| 5 1 out of 4
_____ | 6 1 out of 5
_____ | 7 2 out of 5
_____ | 8 9 out of 10
_____ |
| 9 23 out of 25
_____ | 10 18 out of 50
_____ | 11 3 out of 5
_____ | 12 8 out of 25
_____ |
| 13 2 out of 10
_____ | 14 6 out of 20
_____ | 15 11 out of 20
_____ | 16 28 out of 50
_____ |

YOU DO

On your own: **Easy** keep going: Round 2.

16 QUESTIONS MORE

ROUND 2

Questions 17–32: write the first amount as a fraction of the second, in simplest form.

- | | | | |
|------------------------|------------------------|------------------------|------------------------|
| 17 6 out of 20 | 18 5 out of 16 | 19 15 out of 30 | 20 3 out of 20 |
| 21 17 out of 30 | 22 11 out of 24 | 23 58 out of 60 | 24 11 out of 20 |
| 25 13 out of 24 | 26 22 out of 40 | 27 44 out of 60 | 28 11 out of 12 |
| 29 1 out of 16 | 30 13 out of 16 | 31 18 out of 40 | 32 4 out of 12 |

 Level 2 · Building up

Medium · 20 questions

WE DO

With your teacher: copy what they write on the board.

Example 2

Change to the same units, then write as a fraction and a percentage.

- a** Write 300 m as a fraction of 1.5 km. **b** Write 300 m as a percentage of 1.5 km.

Your teacher works this out. Copy every step.

Your teacher works this out. Copy every step.

YOU DO

On your own: **Medium** practise the skill. Work down the page, one round at a time. Write each answer on its line.

20 QUESTIONS

REMEMBER

Use the same units for both.

ROUND 1

Questions 1–12: change to the same units, then write as a percentage.

- | | | |
|------------------|------------------|-----------------|
| 1 200 g of 4 kg | 2 50 c of 2 \$ | 3 10 c of 1 \$ |
| 4 10 c of 3 \$ | 5 100 m of 4 km | 6 30 cm of 2 m |
| 7 200 m of 1 km | 8 500 m of 1 km | 9 18 min of 3 h |
| 10 750 g of 2 kg | 11 15 min of 4 h | 12 25 cm of 1 m |

YOU DO On your own: **Medium** keep going: Round 2.

8 QUESTIONS MORE

ROUND 2 Questions 13–20: test scores: write each as a percentage.

13 20 / 25

14 16 / 20

15 15 / 20

16 17 / 40

17 39 / 40

18 14 / 40

19 31 / 50

20 34 / 40

Level 3 · Challenge

Challenging · 3 questions · then try the extension

WE DO With your teacher: copy what they write on the board.

Example 3 Write as a percentage, then compare.

a Write 45 min as a percentage of 2 hours.

Your teacher works this out. Copy every step.

b Mia scored 34/40 and Jake 42/50. Who did better?

Your teacher works this out. Copy every step.

YOU DO On your own: **Challenging** Problems.

3 QUESTIONS

REMEMBER Simplify the fraction.

1 Mia scored 34/40 and Jake scored 42/50. Who did better?

2 A team won 18 of 24 games. What percentage did they lose?

3 A \$120 price dropped to \$90. What is the percentage decrease?

✕ Show off your skill: tear along this line and hand it to your teacher ✕

YOU DO **Show off your skill**

Choose **ONE** question. Show your working.

Name _____ Class _____

Level 1

Write 12 out of 16 as a fraction.

Level 2

Write 21 out of 25 as a percentage.

Level 3

Write 40 cm as a percentage of 2 m.

Extension · Stretch yourself**Worked example**

A shirt that cost \$40 now costs \$46. Find the percentage increase.

1. Increase = \$6
2. $\frac{6}{40} \times 100\%$

Answer: 15%

YOU DO

On your own: read the worked example, then try these.

- E1** A town grew from 2500 to 3100 people. Find the percentage increase.

- E2** A \$75 game is on sale for \$60. What percentage discount is this?

Lesson summary

Missed the lesson? Start here.

- Remember**
1. Write **part ÷ whole** as a fraction.
 2. Use the same units for both.
 3. Simplify the fraction.
 4. Multiply by 100% for a percentage.

Level 1

51 out of 60

1. $\frac{51}{60}$
2. $\times 100\%$

Answer: 85%

Level 2

300 m of 1.5 km

1. 1.5 km = 1500 m
2. $\frac{300}{1500}$

Answer: $\frac{1}{5} = 20\%$

Level 3

45 min of 2 h

1. $\frac{45}{120}$

Answer: $\frac{3}{8} = 37.5\%$

Extension

\$40 → \$46

1. $\frac{6}{40} \times 100\%$

Answer: 15%

This space is left blank on purpose. It is the back of your exit ticket.